

3.3 Find the Laplace $X(s)$ given that $x(t)$ is

(a) $2tu(t-4)$

(b) $5\cos t \delta(t-2)$

(c) $e^{-t}u(t-\tau)$

(d) $\sin 2t u(t-\tau)$

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$$(a) \mathcal{L}[2tu(t-4)]$$

$$= -2 \frac{d}{ds} \left(\frac{1}{s} e^{-4s} \right)$$

$$= -2 \frac{-4se^{-4s} - e^{-4s}}{s^2}$$

$$= \frac{8se^{-4s} + 2e^{-4s}}{s^2} *$$

$$(b) \mathcal{L}[5\cos t \delta(t-2)]$$

$$= \mathcal{L}[5\cos(2)\delta(t-2)]$$

$$= 5\cos(2)e^{-2s} *$$

$$(c) \mathcal{L}[e^{-t}u(t-\tau)]$$

$$= \mathcal{L}[e^{-\tau} e^{-(t-\tau)} u(t-\tau)]$$

$$= e^{-\tau} \cdot e^{-\tau s} \cdot \frac{1}{s+1}$$

$$= \frac{e^{-\tau(s+1)}}{s+1} *$$

$$(d) \mathcal{L}[\sin 2t u(t-\tau)]$$

$$= \mathcal{L}[\sin(2(t-\tau) + 2\tau) u(t-\tau)]$$

$$= \mathcal{L}[(\sin 2(t-\tau) \cos 2\tau + \cos 2(t-\tau) \sin 2\tau) u(t-\tau)]$$

$$= \cos 2\tau e^{-\tau s} \frac{2}{s^2 + 4} + \sin 2\tau e^{-\tau s} \frac{s}{s^2 + 4}$$

$$= e^{-\tau s} \frac{2\cos 2\tau + s\sin 2\tau}{s^2 + 4} *$$

3.13 Find the inverse Laplace transform for

$$(a) X(s) = \frac{3s+1}{s^2+2s+5}$$

$$(b) Y(s) = \frac{3s+7}{s^2+3s+2}$$

$$(c) Z(s) = \frac{s^2-8}{s^2-4}$$

$$(d) H(s) = \frac{12}{(s+2)^2}$$

$$\begin{aligned} (a) X(s) &= \frac{3s+1}{s^2+2s+5} \\ &= \frac{3(s+1)}{(s+1)^2+2^2} - \frac{2}{(s+1)^2+2^2} \end{aligned}$$

$$x(t) = e^{-t} (3 \cos 2t - \sin 2t) u(t) \neq$$

$$\begin{aligned} (b) Y(s) &= \frac{3s+7}{s^2+3s+2} \\ &= \frac{A}{(s+2)} + \frac{B}{(s+1)} \end{aligned}$$

$$A = (s+2)Y(s)|_{-2} = \frac{1}{-1} = -1$$

$$B = (s+1)Y(s)|_{-1} = 4$$

$$Y(s) = \frac{-1}{s+2} + \frac{4}{s+1}$$

$$y(t) = (-e^{-2t} + 4e^{-t}) u(t) \neq$$

$$(c) Z(s) = \frac{s^2-8}{s^2-4}$$

$$= 1 + \frac{-4}{s^2-4}$$

$$= 1 + \frac{A}{(s+2)} + \frac{B}{(s-2)}$$

$$A = (s+2) \frac{-4}{(s^2-4)} \Big|_{-2} = 1$$

$$B = (s-2) \frac{-4}{(s^2-4)} \Big|_2 = -1$$

$$Z(s) = 1 + \frac{1}{s+2} + \frac{-1}{s-2}$$

$$z(t) = (\delta(t) + e^{-2t} - e^{2t}) u(t) \neq$$

$$(d) H(s) = \frac{12}{(s+2)^2}$$

$$\mathcal{L}[t u(t)] = -\frac{d}{ds} \frac{1}{s} = \frac{1}{s^2}$$

$$h(t) = 12 t e^{-2t} u(t) \neq$$

3.14 Find the inverse Laplace transform of the following functions:

(a) $F(s) = \frac{20(s+2)}{s(s^2+6s+25)}$

(b) $P(s) = \frac{6s^2+36s+20}{(s+1)(s+2)(s+3)}$

$$(a) F(s) = \frac{20(s+2)}{s(s^2+6s+25)}$$

$$= \frac{A}{s} + \frac{Bs+C}{(s+3)^2+4^2}$$

$$A = s F(s) \Big|_0 = \frac{40}{25} = 1.6$$

$$1.6(s^2+6s+25) + s(Bs+C) = 20(s+2)$$

$$s^2: 1.6 + B = 0 \Rightarrow B = -1.6$$

$$s^1: 9.6 + C = 20 \Rightarrow C = 10.4$$

$$F(s) = \frac{1.6}{s} + \frac{-1.6s+10.4}{(s+3)^2+4^2} \quad \frac{15.2}{4}$$

$$= \frac{1.6}{s} + \frac{-1.6(s+3)}{(s+3)^2+4^2} + \frac{3.8 \cdot 4}{(s+3)^2+4^2}$$

$$f(s) = (1.6 - 1.6 e^{-3t} \cos 4t + 3.8 e^{-3t} \sin 4t) u(t) \neq$$

$$\begin{aligned} \text{(b)} \quad P(s) &= \frac{6s^2 + 36s + 20}{(s+1)(s+2)(s+3)} \\ &= \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} \end{aligned}$$

$$A = (s+1)P(s)|_{-1} = \frac{-10}{2} = -5$$

$$B = (s+2)P(s)|_{-2} = \frac{-28}{-1} = 28$$

$$C = (s+3)P(s)|_{-3} = \frac{-34}{2} = -17$$

$$P(s) = \frac{-5}{s+1} + \frac{28}{s+2} + \frac{-17}{s+3}$$

$$p(t) = (-5e^{-t} + 28e^{-2t} - 17e^{-3t})u(t) \quad \#$$

3.18 Let $F(s) = \frac{5(s+1)}{(s+2)(s+3)}$

- (a) Use the initial and final value theorems to find $f(0)$ and $f(\infty)$.
 (b) Verify your answer in part (a) by finding $f(t)$ using partial fractions.

$$\begin{aligned} \text{(a)} \quad f(0) &= \lim_{s \rightarrow \infty} sF(s) \\ &= \lim_{s \rightarrow \infty} \frac{s \cdot 5(s+1)}{(s+2)(s+3)} \\ &= \lim_{s \rightarrow \infty} \frac{5s^2 + 5s}{s^2 + 5s + 6} \\ &= 5 \end{aligned}$$

$$\begin{aligned} f(\infty) &= \lim_{s \rightarrow 0} sF(s) \\ &= \lim_{s \rightarrow 0} \frac{s \cdot 5(s+1)}{(s+2)(s+3)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad F(s) &= \frac{5s+5}{(s+2)(s+3)} \\ &= \frac{A}{s+2} + \frac{B}{s+3} \end{aligned}$$

$$A = (s+2)F(s)|_{s=-2} = -5$$

$$B = (s+3)F(s)|_{s=-3} = 10$$

$$F(s) = \frac{-5}{s+2} + \frac{10}{s+3}$$

$$f(t) = (-5e^{-2t} + 10e^{-3t})u(t)$$

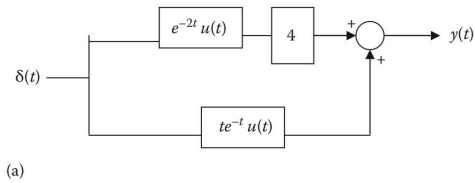
$$f(0) = -5 + 10 = 5$$

$$f(\infty) = 0 - 0 = 0$$

3.26 For each of the systems shown in Figure 3.28, use Laplace transform to find $y(t)$.

$$\begin{aligned} \text{(a)} \quad Y(s) &= \mathcal{L}[\delta(t)] \cdot (\mathcal{L}[e^{-2t}u(t)] \cdot \mathcal{L}[4] + \mathcal{L}[te^{-t}u(t)]) \\ &= 1 \cdot \left(\frac{4}{s+2} + \frac{1}{(s+1)^2} \right) \end{aligned}$$

$$y(t) = (4e^{-2t} + te^{-t})u(t)$$



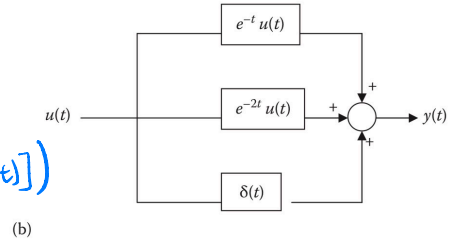
$$\begin{aligned} \text{(b)} \quad Y(s) &= \mathcal{L}[u(t)] \cdot (\mathcal{L}[e^{-t}u(t)] + \mathcal{L}[e^{-2t}u(t)] + \mathcal{L}[\delta(t)]) \\ &= \frac{1}{s} \cdot \left(\frac{1}{s+1} + \frac{1}{s+2} + 1 \right) \end{aligned}$$

$$= \frac{1}{s} \cdot \left(\frac{1}{s+1} + \frac{1}{s+2} + 1 \right)$$

$$= \frac{1}{s} - \frac{1}{s+1} + 0.5 \left(\frac{1}{s} - \frac{1}{s+2} \right) + \frac{1}{s}$$

$$= 2.5 \frac{1}{s} - \frac{1}{s+1} - \frac{0.5}{s+2}$$

$$y(t) = (2.5 - e^{-t} - 0.5e^{-2t})u(t)$$



3.43 An LTI system is described by

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} - 3x(t)$$

(a) Determine the transfer function of the system.

(b) Obtain the impulse response of the system.

$$(a) \quad x(0) = y(0) = y'(0) = 0$$

$$\xrightarrow{\text{transfer}} (s^2 Y(s) - s y(0) - y'(0)) + 2(sY(s) - y(0)) + 2Y(s) = sX(s) - x(0) - 3X(s)$$

$$(s^2 + 2s + 2)Y(s) = (s - 3)X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s - 3}{s^2 + 2s + 2}$$

$$(b) \quad H(s)$$

$$= \frac{s - 3}{s^2 + 2s + 2}$$

$$= \frac{s + 1}{(s + 1)^2 + 1} + \frac{-4}{(s + 1)^2 + 1}$$

$$h(t) = (e^{-t} \cos t - 4e^{-t} \sin t) u(t) *$$

3.46 Consider the function

$$H(s) = \frac{s^2 + 6s + 10}{s^3 + 7s^2 + 11s + 5}$$

Use the MATLAB residue function to obtain the inverse Laplace transform of $H(s)$.

```
>> HW3_3_46
```

```
r =
```

```
0.3125
```

```
0.6875
```

```
1.2500
```

```
p =
```

```
-5.0000
```

```
-1.0000
```

```
-1.0000
```

```
k =
```

```
[]
```

$$H(s) = \frac{0.3125}{s+5} + \frac{0.6875}{s+1} + \frac{1.25}{(s+1)^2}$$

$$h(s) = (0.3125 e^{-5t} + 0.6875 e^{-t} + 1.25 t e^{-t}) u(t) \quad \#$$

3.49 A linear system is represented by its transfer function

$$H(s) = \frac{2s + 5}{(s + 2)(s + 3)}$$

Use MATLAB to determine and plot:

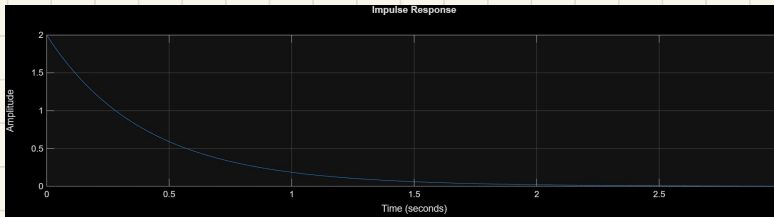
- (a) The impulse response
- (b) The step response.

(a) $h(t) = (e^{-2t} + e^{-3t})u(t)$

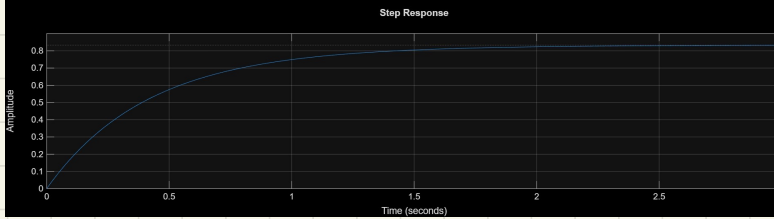
(b) $y(t) = \left(\frac{5}{6} - \frac{1}{2}e^{-2t} - \frac{1}{3}e^{-3t}\right)u(t)$

```
1. 脈衝響應數學式:  
h(t) = ( 1.000*e^(-3.0t) + 1.000*e^(-2.0t) ) * u(t)  
  
2. 階躍響應數學式:  
y(t) = ( 0.833 - 0.500*e^(-2.0t) - 0.333*e^(-3.0t) ) * u(t)
```

(a)



(b)



3.51 Use MATLAB to find the zeros and poles of these functions:

(a) $\frac{s-2}{(s+1)^2+9}$

(b) $\frac{s^2+2s+5}{s(s^2+4s+13)}$

(c) $\frac{s^2+10s+5}{s^3+4s^2+10s+6}$

```
>> HW3_3_51_a
```

```
z_a =
```

```
2
```

```
p_a =
```

```
-1.0000 + 3.0000i  
-1.0000 - 3.0000i
```

```
k_a =
```

```
1
```

```
>> HW3_3_51_b
```

```
z_b =
```

```
-1.0000 + 2.0000i  
-1.0000 - 2.0000i
```

```
p_b =
```

```
0.0000 + 0.0000i  
-2.0000 + 3.0000i  
-2.0000 - 3.0000i
```

```
k_b =
```

```
1
```

```
>> HW3_3_51_c
```

```
z_c =
```

```
-9.4721  
-0.5279
```

```
p_c =
```

```
-1.5956 + 2.2075i  
-1.5956 - 2.2075i  
-0.8087 + 0.0000i
```

```
k_c =
```

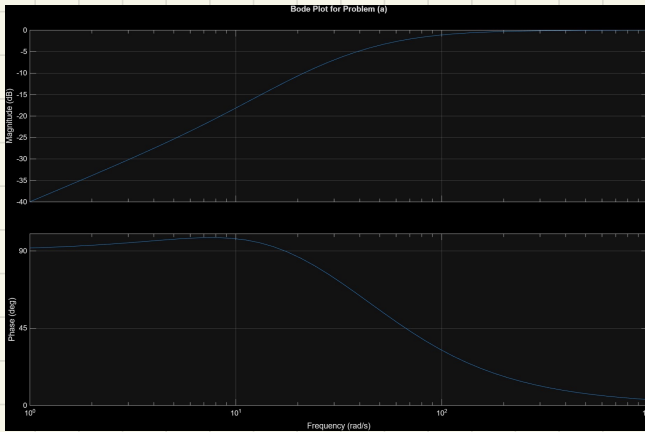
```
1
```

3.53 Obtain the Bode plots for the following transfer functions using MATLAB:

(a) $H(s) = \frac{s(s+10)}{(s+20)(s+50)}$

(b) $H(s) = \frac{s+1}{(s+2)(s^2+22.5s+16)}$

(a)



(b)

